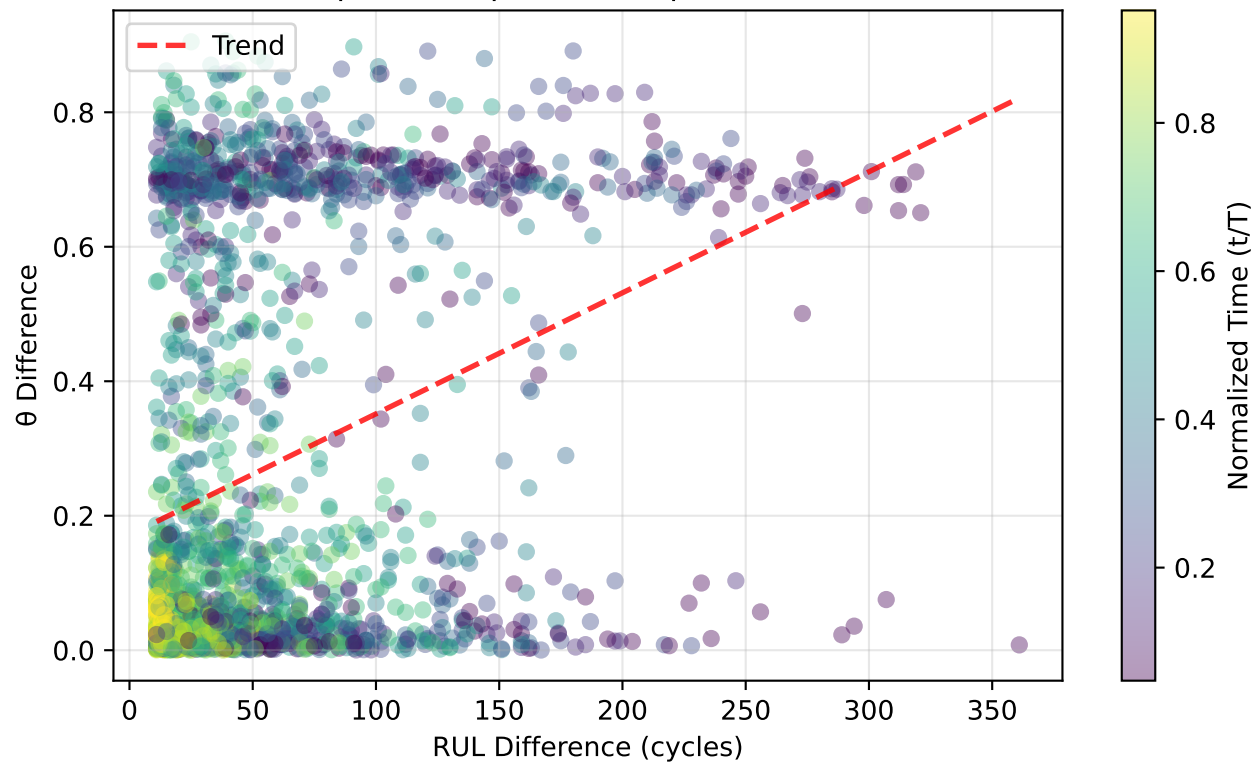
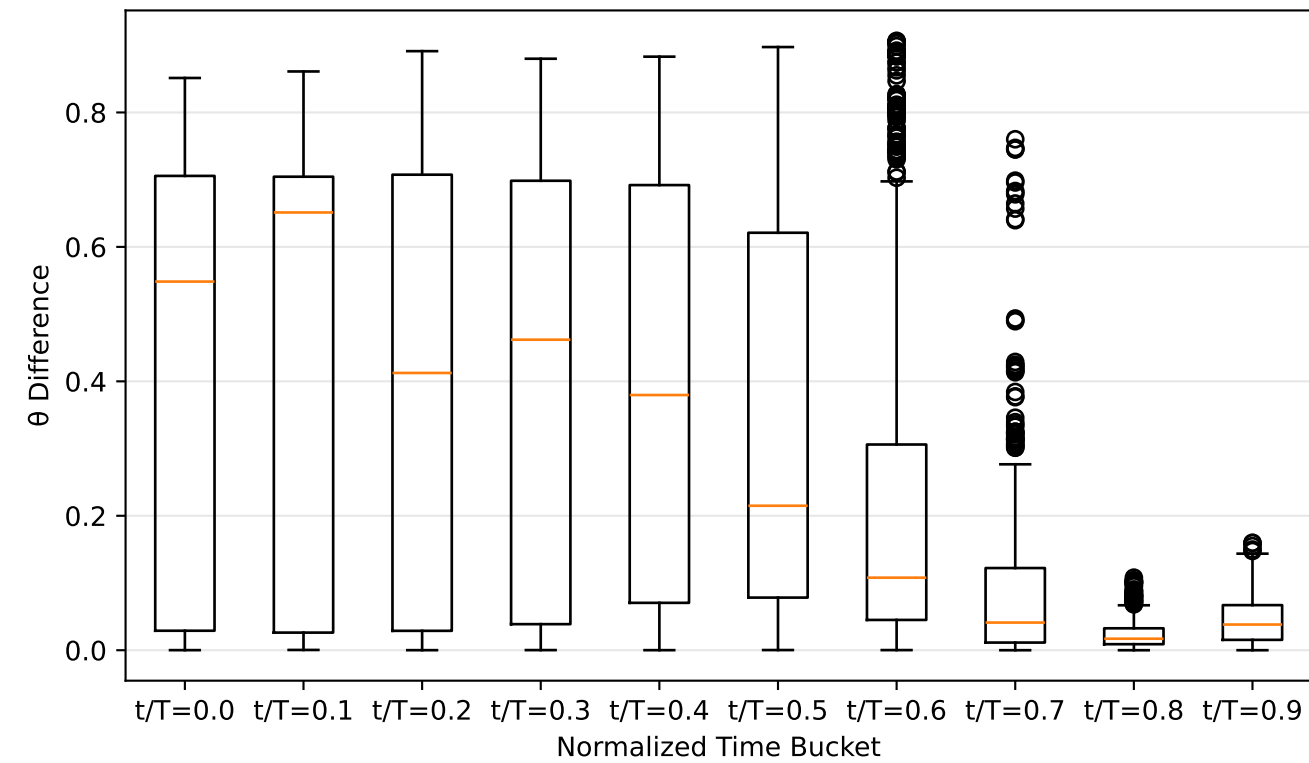


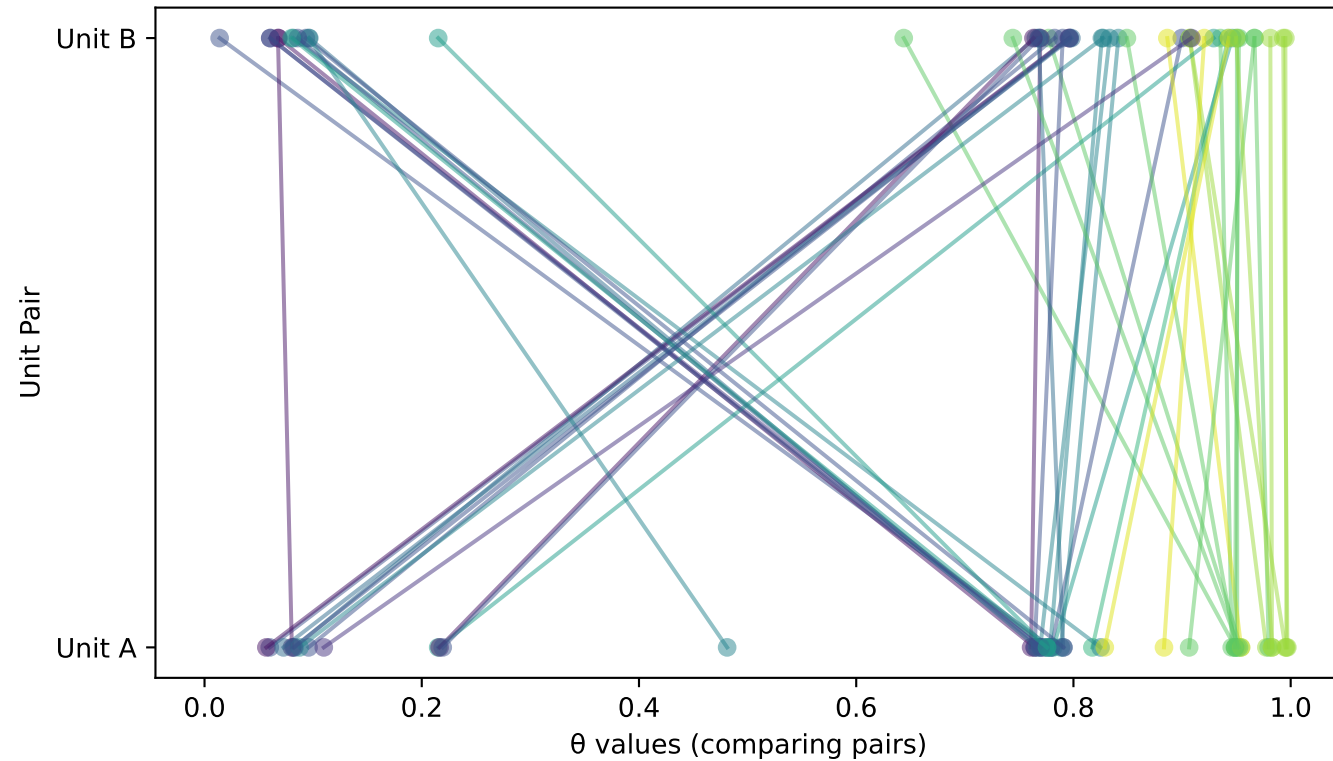
Health State Difference vs θ Difference
(Spearman $\rho=0.2922$, $p=4.95e-99$)



θ Difference Across Different Times



Sample Pairs: Showing θ Differences Within Pairs



Test 1c: Same-Time Different-Health Analysis

Dataset: FD003
Latent Dim: 4
Window Size: 10

Results:

Comparison Pairs: 5000

RUL Difference:
Mean: 55.9 cycles
Median: 35.0 cycles
Max: 361.0 cycles

θ Difference:
Mean: 0.2722
Median: 0.0880
Max: 0.9064

Correlation:
Spearman $\rho = 0.2922$
 p -value = $4.95e-99$

Interpretation:

If $\rho \gg 0$ and $p < 0.05$:
→ θ TRACKS STATE (different health leads to different θ even at same t/T)

If $\rho \approx 0$ or $p > 0.05$:
→ θ may be learning just t/T (health difference doesn't affect θ)